



The Widget Wizards

Your homescreens look bare without widgets, but which ones of the thousands available should you use? Let us choose for you, and also teach you how to make your own.

<http://dgit.in/WdgtWzrd>

recovery environment – all other recoveries are controlled by a device's volume and power button. TWRP even supports themes. It is also used to make backups and flash ROMs on your Android devices.

Essentially, to carry out a TWRP recovery, you need to follow these steps:

- Enable USB debugging.
- Download the TWRP recovery image for your phone and place it in C:\
- Boot into your phone's bootloader and connect it to PC.
- Wait until you see 'fastboot USB' on your screen.
- Launch command prompt on your computer and run this command:
`fastboot flash recovery c:\<name of image>.img`
- After the process is finished, go to bootloader again – you should see the TWRP custom recovery.
- Select Backup option and choose the relevant categories before starting the process
- Once successfully completed, you should have a TWRP custom recovery image on your storage
- The 'Restore' option on the first menu takes you through the recovery process

ROOTING YOUR PHONE

Now it's time for the real deal – rooting your phone. There are several ways to do this and you will need to decide which one's best for you. Before you begin, make sure your device is fully charged – you do not want your device dying on you mid-root, trust us!

You need to have both USB Debugging and OEM Unlocking enabled on your device – both of which are available under Developer options. Now, just in case you haven't enabled Developer options yet, here's a quick few steps on how to do it all in one go:

- Go to Settings and scroll down to the very bottom then tap on 'About Phone'.
- Scroll until you find 'Build number' and the tap on it seven times. The Developer Options section should now be available on the main Settings window.
- Under Developer Options, enable USB Debugging and OEM Unlocking



There was a time when the process was quite complex

INSTALL ANDROID SDK TOOLS

- Get the SDK tools from <http://dgit.in/AndrSDK> and install them. Remember the installed directory.
- Launch the SDK tools and uncheck everything except Android SDK Platform-tools at the top.



Make your own smartphone case

When just buying one is too mainstream, real geeks make their own. Would you like to get a Dbrand-like carbon-fiber look for one-tenth the price? Maybe you want a spray-on Rubber case? You can have it all by going to

<http://dgit.in/DIYCases>

- Click on Install 2 packages at the bottom right. Check Accept license and click Install.

CHECK COMPATIBILITY

Before starting the process, make sure your desktop/laptop detects the phone correctly when connected via USB. If not, just do a quick Google search for the drivers from your manufacturer and install them.

UNLOCK BOOTLOADER

Another thing that needs to be done before you get started is the unlocking of your bootloader. The bootloader is basically the program that controls your device's startup process. It has control over what applications run during that process. Hence, unlocking it is a key to rooting your device.

This is where things get manufacturer specific. Some manufacturers require you to get a key to unlock the bootloader – something that they allow only registered developer accounts to do. Usually, for this, you have to put your device into fastboot mode – something that again differs depending on the manufacturer. The most common way to do this is to reboot and hold the power and volume down buttons for 10 seconds, but it could differ for your phone.

Once in fastboot, open your computer's command prompt. If your device requires a code this is the stage where you'll see a long string of characters, that you will need to provide on the manufacturer's website to get a key and further instructions on unlocking the bootloader. Otherwise, this is the process you need to follow.

- Connect it to your computer and bring it into fastboot again.
- Bring up the command prompt.

For Google Nexus and Pixel devices, the commands are easy:

- Nexus phones: Type "`fastboot oem unlock`" (without quotes) and hit Enter
- Pixel phones: Type "`fastboot flashing unlock`" (without quotes) and hit Enter

Motorola's command is a little different:

- Type "`oem unlock UNIQUE_KEY`" (without quotes), replacing "UNIQUE_KEY" with the code you received